

CHIEN-YI YANG

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Objective

5th-year CSE Ph.D. candidate at UCSD (expected graduation Dec. 2026), seeking full-time AI compute architect roles. Focused on software-hardware co-design for AI/ML accelerators, parameterized architecture modeling, analytical PPA engines, and tiling/scheduling optimization for tensor, transformer, and LLM workloads.

Education

University of California San Diego

Sept. 2022 – Dec. 2026 (Expected)

Ph.D. in Computer Science and Engineering

National Taiwan University

Sept. 2017 – June 2021

B.S. in Electrical Engineering

Research Experience

University of California San Diego

Sept. 2022 – Present

Graduate Research Assistant

- **PNMAX: Mapping-Architecture Co-Exploration for Processing-near-Memory AI Accelerators (MICRO'26, First Author):** End-to-end DSE framework for processing-near-memory AI accelerators. Built a parameterized graph-based architecture representation, a hardware-aware mapping formulation, and a bottom-up analytical PPA engine over processing units, buffers, DRAM banks, and interconnects. Achieved up to 32.1x speedup over cycle-accurate simulators with $R^2 > 0.97$, and discovered Pareto-optimal architecture-mapping design points for **transformer/LLM workloads**.
- **OptiPIM: MILP-Based DNN Inference Acceleration:** Co-developed a mixed-integer linear programming engine maximizing DNN inference throughput on digital processing in memory (PIM); outperformed the deployed Samsung HBM-PIM heuristic mapping by **72.8x** and prior mapping methods by $\geq 1.9x$. **ISCA'25 Best Artifact Award**.
- **Multi-Objective Co-Optimization for Analog PIM:** Noise-aware Bayesian optimization (**BoTorch**) jointly tuning software and hardware parameters under device non-idealities. Achieved **3.3x** higher constraint satisfaction, **4.28%** accuracy gain, **35.4%** power reduction, and **49x** speedup.
- **NN-Steiner: ML for EDA :** Mixed neural-algorithmic solver for the rectilinear Steiner minimum tree problem; built a **PyTorch Distributed** training system at **>98% GPU utilization**, with up to **2.99%** improvement over state-of-the-art.

Industry Experience

MediaTek Research

Dec. 2021 – June 2022

R&D Engineer Intern

Taipei, Taiwan

- **Distributed LLM Training Infrastructure:** Deployed a large-scale distributed training pipeline (data + model parallelism) for a T5-based LLM on a cluster of **100+ GPUs** via **PyTorch Distributed**. Achieved **30x** end-to-end throughput speedup through **PyTorch Profiler**-based critical-path optimization across compute, collective communication, and data loading.
- **RL-Based Floorplanning:** Designed an RL methodology for chip floorplanning, accelerating the design flow **>3x**.

Selected Publications

- **C.-Y. Yang**, J. Liu, Y. Pan, T. Rosing, M. Zhou, L. Josipovic. PNMAX: Mapping-Architecture Co-Exploration Framework for Processing near Memory. **MICRO'26**.
- **C.-Y. Yang**, F. Ponzina, M. Zhou, R. Ayoub, P. Mercati, T. Rosing. Constraint-Scheduled Bayesian Optimization for Software-Hardware Co-Optimization on HDnn-PIM. **ESL'26**.
- E. Singh, D. Wu, **C.-Y. Yang**, T. Rosing, R. Yu, Y. Ma. Divide and Learn: Multi-Objective Combinatorial Optimization at Scale. **ICML'26**.
- J. Liu, M. Zhou, Y. Pan, **C.-Y. Yang**, L. Josipovic, T. Rosing. OptiPIM: Optimizing Processing-in-Memory Acceleration Using Integer Linear Programming. **ISCA'25**. [**ISCA Best Artifact Award**]
- **C.-Y. Yang**, M. Zhou, F. Ponzina, S. S. Prakash, R. Ayoub, P. Mercati, M. Subedar, T. Rosing. Multi-objective software-hardware co-optimization for HD-PIM via noise-aware Bayesian optimization. **ICCAD'24**.
- Kahng, A. B., R. R. Nerem, Y. Wang, and **C.-Y. Yang**. NN-Steiner: A mixed neural-algorithmic approach for the rectilinear Steiner minimum tree problem. **AAAI'24**. (**Alphabetical Order**)

Skills

Languages & Systems: C/C++, Python, Rust, Tcl; PyTorch, PyTorch Distributed

Architecture & Performance Modeling: Analytical PPA modeling, roofline analysis, Timeloop, CiMLoop, CACTI, Accelergy-style cost models; cycle-accurate simulation methodologies

EDA: Design Compiler, Innovus, RTL synthesis & physical design